GeoODE-KD: Distilling Sentence Embeddings as Teacher-Guided Geometric Dynamics

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Abstract

Table 1: Main results for distillation from Qwen3-Embedding-4B to BERT-base. Classification datasets are evaluated with F1, pair-classification datasets with average precision (AP), and semantic textual similarity datasets with Spearman correlation. Higher is better. Among non-teacher models, the best result is shown in bold and the second-best result is underlined.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			
Method	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	Avg.
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)										
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	54.36
Stella	91.08	73.40	64.40	<u>86.23</u>	83.22	<u>68.98</u>	78.38	71.31	79.16	77.35
CDM	89.71	70.76	52.67	84.03	71.47	**69.83**	71.36	68.87	74.68	72.60
DSKD	88.89	69.31	53.39	**86.35**	67.73	68.89	70.58	67.99	74.29	71.94
EMO	85.92	69.39	55.41	85.79	64.34	67.73	63.79	64.41	70.14	69.60
TALAS	**93.34**	**74.45**	**72.70**	86.16	**87.78**	68.11	<u>80.70</u>	**77.34**	**83.57**	**80.46**
GeoODE-KD	<u>92.92</u>	<u>74.05</u>	<u>72.49</u>	85.89	<u>86.12</u>	66.60	**80.78**	<u>76.01</u>	<u>81.23</u>	<u>79.57</u>